

MUHAMMAD ANAS

Python Developer — Building RAG Pipelines, REST APIs & Data Systems

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PROFESSIONAL SUMMARY

Python developer who ships complete, working systems. Built a production, Shopify-integrated AI support agent (Gemini embeddings, FAISS vector search, conflict detection, live order lookups) validated by an automated evaluation harness, plus three additional end-to-end applications spanning database design, REST APIs, desktop software, and BI reporting. Comfortable owning a project from schema design to user interface.

TECHNICAL SKILLS

AI / RAG Engineering: Google Gemini API, FAISS vector search, embeddings, LangChain text splitters, prompt design

Machine Learning: Linear Regression, Decision Trees, Random Forest, SVM, k-NN, Naive Bayes, Clustering, PCA

Backend & APIs: Flask, Flask-Login, Flask-WTF, Flask-Limiter, Flask-SQLAlchemy, REST API design, Shopify Admin API

Databases: PostgreSQL, MySQL, SQLite (3NF schema design, joins, indexes, transactions)

Data & Visualization: NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, Power BI (DAX, Power Query)

Frontend & Desktop: HTML, CSS, JavaScript, Jinja2, Tailwind CSS, CustomTkinter, PyInstaller

Security & Tools: CSRF protection, rate limiting, bcrypt hashing, Git, GitHub, Docker, AWS EC2, Nginx

EXPERIENCE

AI Intern — Decode Labs (Remote)

August • 2026

- Built a rule-based chatbot handling intent matching and multi-turn conversational logic
- Trained an Iris species classifier using classical supervised learning, evaluating accuracy across multiple algorithms
- Built a tech-stack recommendation tool matching project requirements to suitable technologies

PROJECTS & DELIVERED SYSTEMS

AI Support Agent — Shopify-Integrated RAG Platform

2026 • Lahore

Flask, Gemini API, FAISS, Shopify Admin API, Nginx

- Built a Shopify-integrated AI agent answering policy Q&A via FAISS retrieval, flagging conflicts
- Built an evaluation harness (Q&A, conflict detection, routing) achieving **98% accuracy across 59 tests**
- Integrated Shopify Admin API for real-time order lookups; rate-limited chat to **10 requests/minute**
- Deployed via systemd behind Nginx with Let's Encrypt SSL; secured admin login with session-based auth

Verve Athletics — Voice-Enabled AI Support Agent

2026 • Lahore

Flask-SocketIO, AssemblyAI, ElevenLabs, Gemini API, FAISS, AWS EC2

- Built a real-time voice agent: AssemblyAI speech-to-text and ElevenLabs speech, streamed live over WebSockets
- Extended a production Gemini/FAISS RAG pipeline with a live voice layer; deployed on AWS EC2

Hotel & Reservation Management System — Full-Stack Web App

2026 • Lahore

Python (Flask REST API), PostgreSQL, HTML/CSS/JavaScript, Docker, AWS EC2

- Designed a 3NF PostgreSQL schema (10 tables) with FKs and CHECK constraints across Lahore & Islamabad branches
- Wrapped reservation and payment creation in one atomic transaction; built a real-time occupancy/revenue dashboard
- Migrated MySQL to PostgreSQL, containerized with Docker, and deployed on AWS EC2 behind Nginx with SSL

StockFlow Pro — Inventory, Sales & Warehouse Management System

2026 • Lahore

Python, CustomTkinter, MySQL

- Shipped a barcode-level warehouse system as a standalone Windows .exe requiring no Python installation
- Implemented a per-carton audit trail with embedded Matplotlib reporting and one-click PDF export via ReportLab

SellerDocs AI — RAG Document Assistant for E-Commerce Sellers

2026 • Lahore

Flask, Gemini API, FAISS, SQLite, Docker

- Built a multi-tenant RAG assistant with per-topic knowledge bases and cross-document conflict detection
- Built an evaluation module tracking retrieval accuracy against test questions with expected sources

EDUCATION

BS Data Science — University of Management & Technology (UMT), Lahore

2024 – Present

Relevant coursework: Machine Learning, Data Analysis (Power BI), Statistics, Database Management, Python Programming

CERTIFICATIONS

- Artificial Intelligence using Python — DigiSkills.pk • Jul 2026
- Data Analytics and Business Intelligence — DigiSkills.pk • Mar 2026
- Data Science & Analytics — HP LIFE • Aug 2025

Languages: Urdu (Native) • English (Fluent)